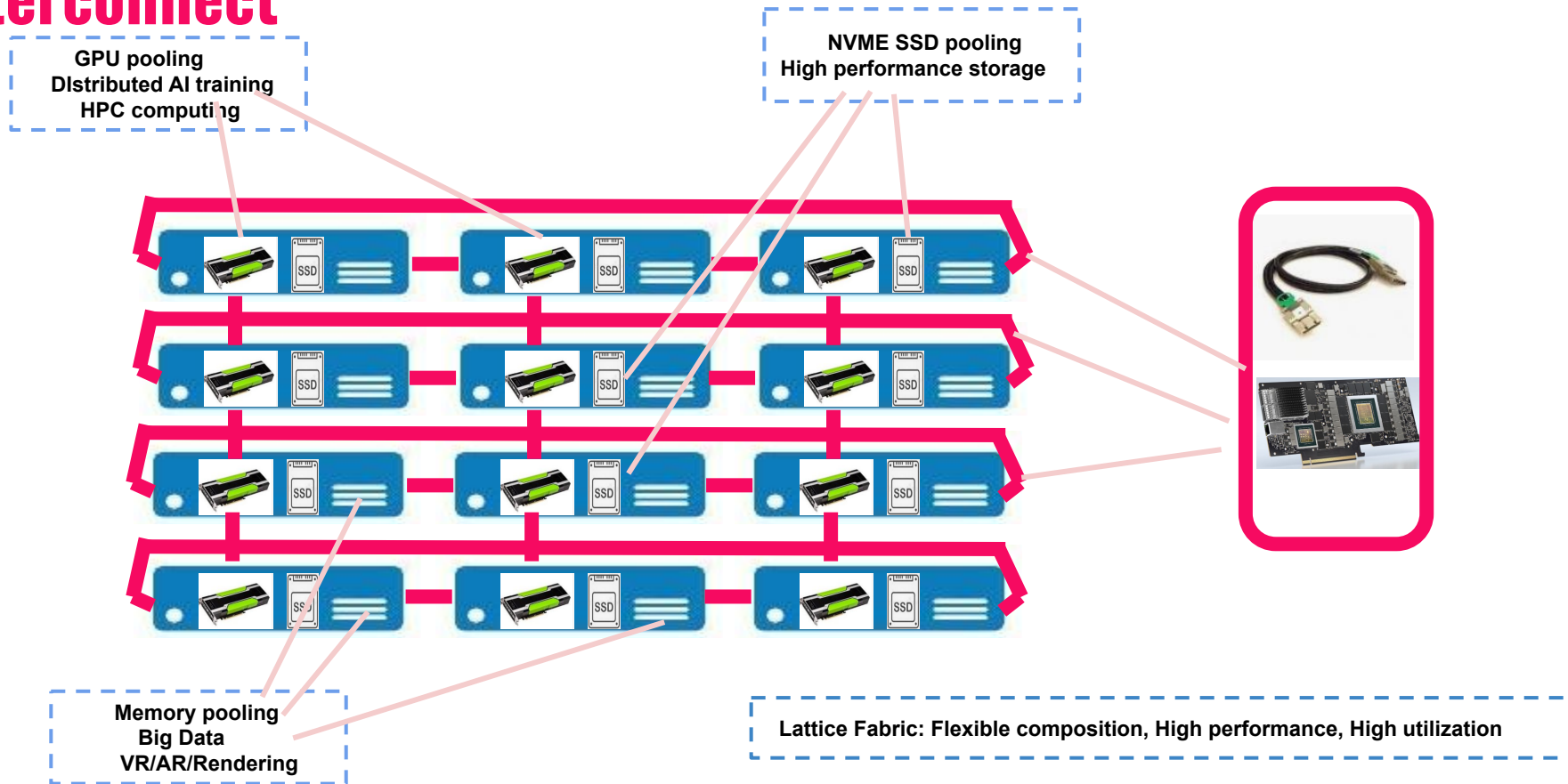


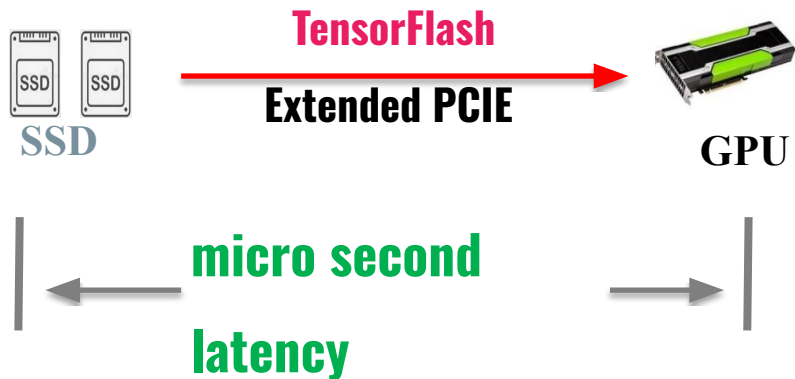
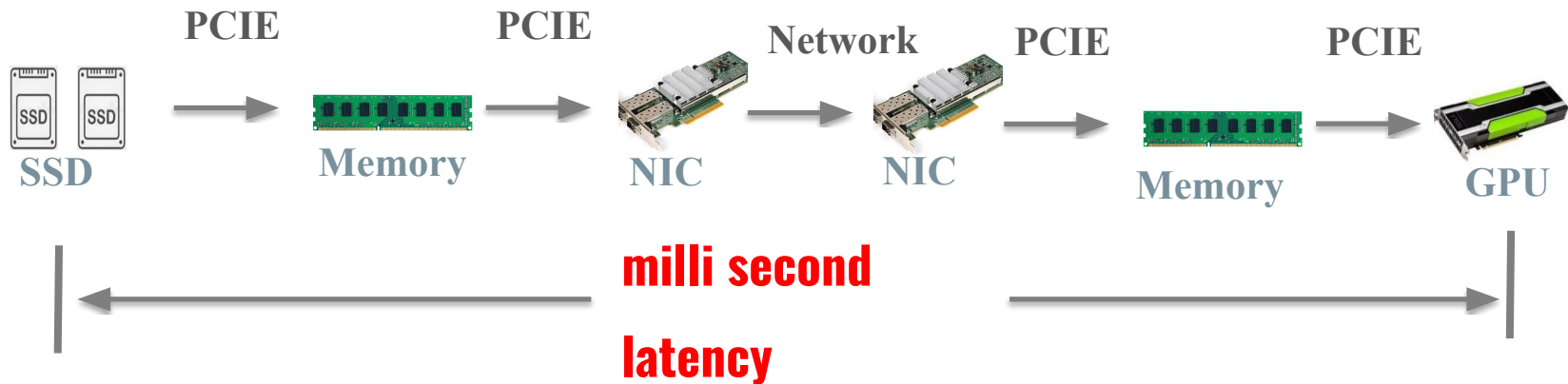
TensorFlash

Democratizing AI Infrastructure

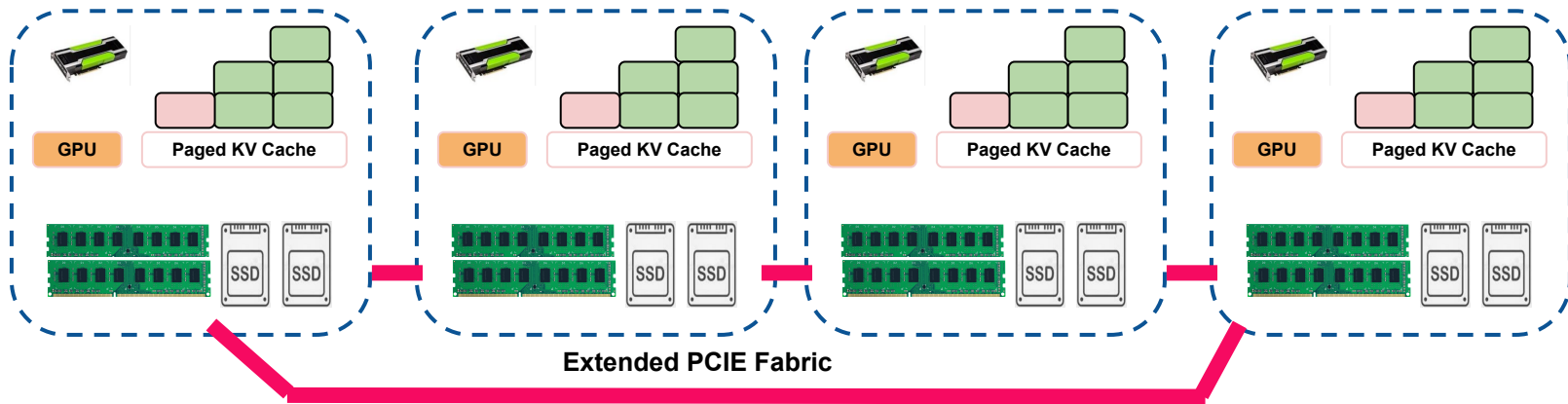
TensorFlash Innovation: Extending PCIe for Server Interconnect



Drastic Reduction in GPU-SSD Data Latency

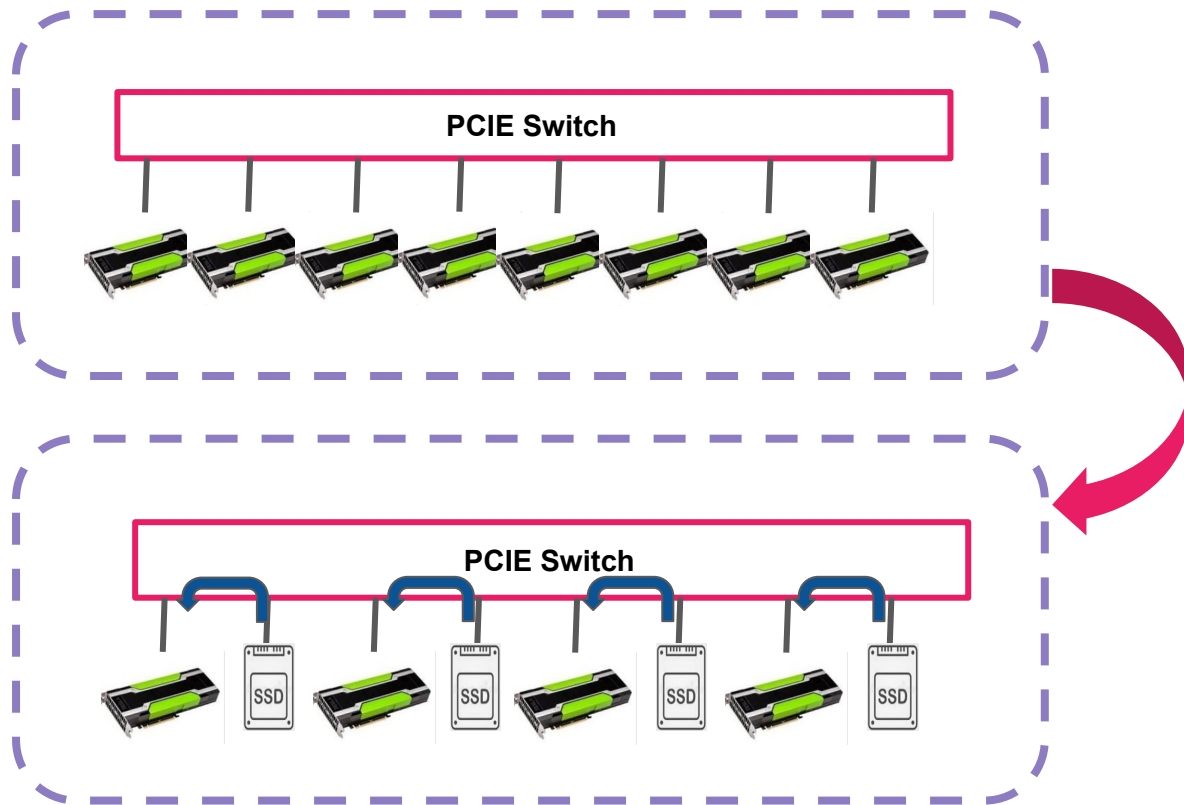


Tightly Coupled GPU/SSD KV Cache for LLM



- Extended PCIE fabric provides pooling for memory and SSD
- Data movement all in hardware, minimal driver/FW involvement
- Drastic improvement in bandwidth and latency compared with traditional RDMA

The Revolutionary TFNative™ Technology

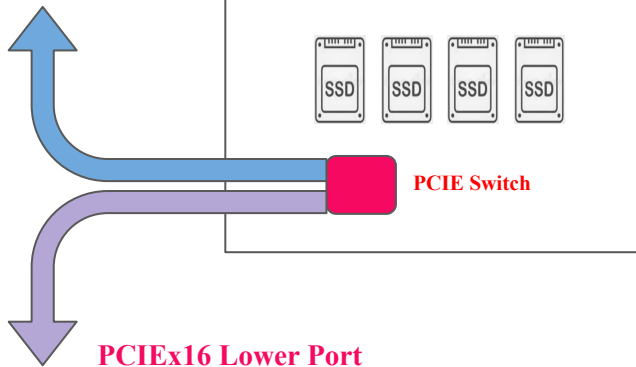


- Each GPU paired with SSD
- GPUs have dedicated computing scratch pad
- SSD hold all model parameters, gradients, optimizer states
- GPU exchange data through neighboring SSD

TensorFlash storage interconnect



PCIEx16 Upper Port



Sever 1



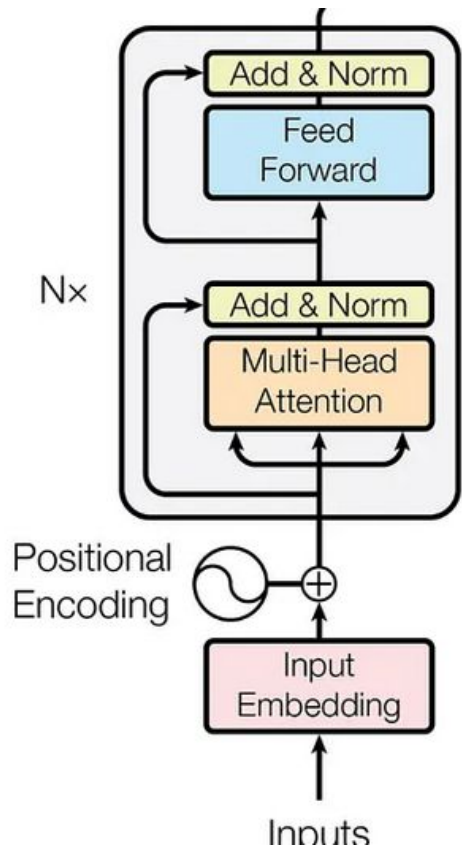
Sever 2



Sever 3



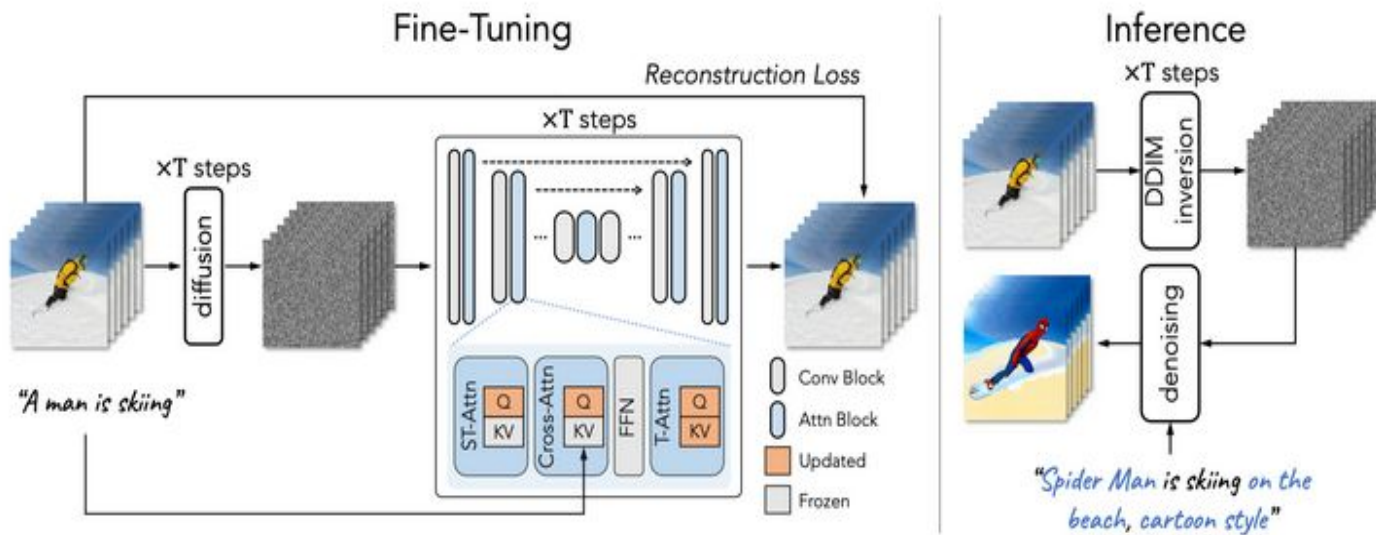
Comparable Performance with Far Fewer GPUs



Case Study of Streaming Computing during LLM Inference

- GPT-3 175B, 96 layer
- Each layer $\sim 1.8\text{B}$, 3.6G Bytes
- Gen6x16 SSD bandwidth $\sim 100\text{GB/s}$
- Model loading time for one layer: $3.6/100 \sim 35\text{ms}$
- Computing: $100\text{K} \times 1.8\text{B} / 1000 \text{ T} \sim 180\text{ms}$

Diffusion Transformer (DiT) - Video Generation

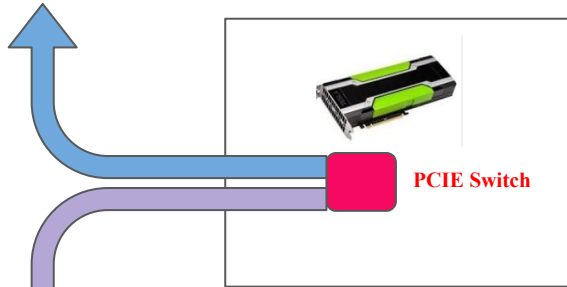


- Both Inverse and Denoise processes have long computing to overlap with IO loading
- High definition video generation requires large HBM in current approach
- TensorFlash provides high quality video generation solution with cost efficient GPU clusters

TensorFlash GPU Interconnect

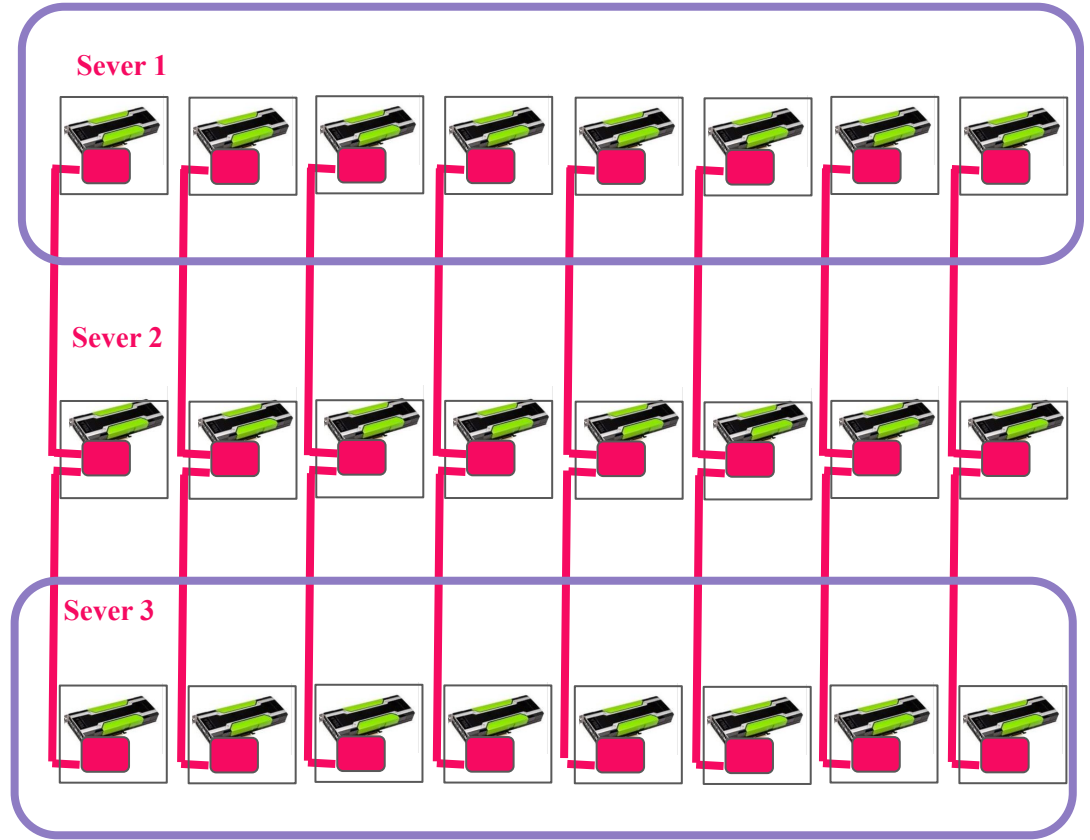


PCIEx16 Upper Port



PCIe Switch

PCIEx16 Lower Port



Large-scale Training with Low-cost GPU or Alternative



- Nvidia 5090 ~ \$2K
- 256 5090 ~\$500K
- Aggregated ~ 50 Pflops
- Train any of the largest LLM
- Multi-Agent Reinforcement Learning

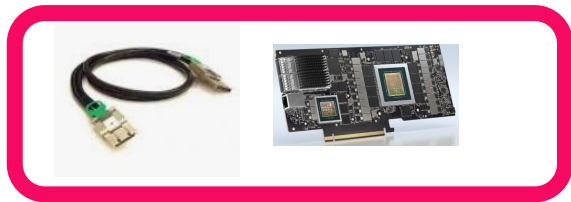
Seamless Integration with All Major Platforms



Hewlett Packard
Enterprise



Lenovo



- Extended PCIe fabric turns any server platform into high performance cluster
- All Flash Array built from standard off-the-shelf components
- Converged and native SSD performance across cluster

GPU/LPU EcoSystem

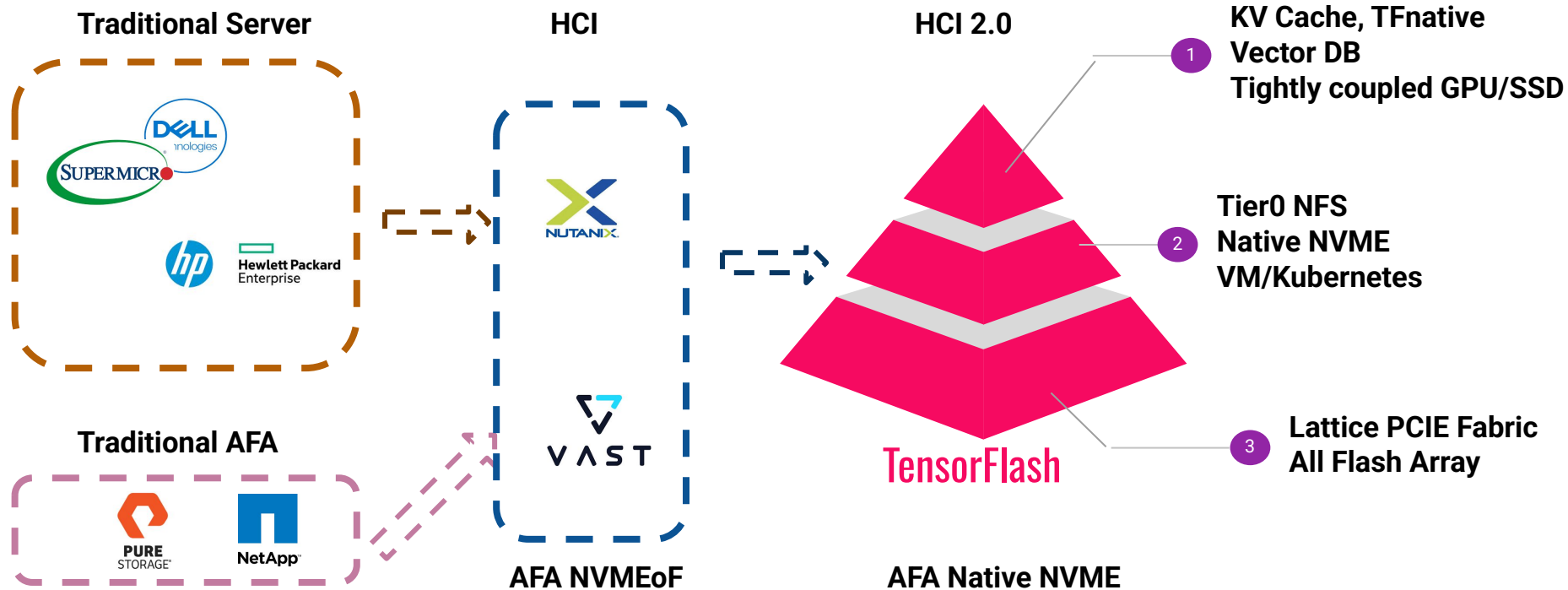


- PCIe is universal interface with all modern GPU/LPU
- TensorFlash naturally provide high bandwidth interconnect fabric

TensorFlash Computing/Price Ratio vs Competitors

GPU	AMD 9700 Intel Arc B79	沐曦	Nvidia H200
Computing Power	190 Tflops	300 Tflops	990 Tflops
Price	10,000 ¥	~55,000 ¥	~210,000 ¥
	1x Tflops 1x Price	1.5x Tflops 5.5x Price	5x Tflops 20x Price

Evolution of Computing Infrastructure



Founding Team

- **Wei Zhou**, worked in Storage industry for over 20 years, have held senior management position at Marvell and SK hynix, leading teams in storage HW/FW/system engineering. Master degree of Applied Physics from Stanford
- **David Chen**, former Vice President and Head of SSD Business at Marvell, founding President of InnoGrit, founding Partner of El Camino Capital. PhD in Electrical Engineering from UC Berkeley
- **K. Wong**, worked in various fields in storage, have broad experience in SSD FW engineering and linux kernel engineering, have held principle engineer positions at Marvell and Sk Hynix
- **X. Hu**, 20-plus years' experience in storage industry, strong kernel and OS engineering expertise, have held various engineering and management positions at Marvell and startups